STEAMROOT · PROJECT PLAN

SteamRoot project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade): workstreams, resources, stage gates (P4 to P6 on hold), decisions requested, reference list

SteamRoot project plan

SteamRoot is at TRL 3 and pipeline stage P3 (design outputs) since the kit 1.9.1 upgrade of 2026-10-25. The next gate, P4 verification, is on hold while the portfolio is capped at TRL 3, and could not start in any case before the written ruling from the Texas boiler authority. Decisions requested from Amish: the concept choice (#14, A and B within 10 %), hood handling (S2), and the open items #12, #13 and #16 to #21 in STR-DEC-001. Method: STANDARDS sections 19 and 23.

> **Safety:** No workstream below includes buying, building or firing anything. The ruling gates all physical work.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Regulatory ruling	Project lead (Amish)	Written ruling from TDLR's boiler program on the open-vented, fired monotube coil (STR-REG-001, R1)	Open prerequisite; not yet requested in writing
Co-design partners	Project lead	One market garden and one nursery near Irving via Texas A&M AgriLife Extension (decided 2026-10-02)	Not yet named
Design (P3)	Design lead	D3 model of both options, drawings, priced BOM, build plan, CAL, STEM check, requirements and traceability	Done 2026-10-25; decisions open in STR-DEC-001
Renders	Coordinator	Options picture (`media/options.png`) and product renders from `cad/src/product_model.py` (hero, studio, exploded) on Amish's workstation	Scenes ready; renders pending
Verification (P4)	Design lead with partners	Soil tests (permeability, skirt leakage, ballast), flue oxygen and temperature, fuel weighing, alarm response, handling trial	On hold (TRL cap 3)
Validation (P5)	Partners	Season of use on a market garden and a nursery	On hold
Commercialization (P6)	None	Only with a named deployment partner	On hold

Resources needed

Table 2. Resources (all for later stages; nothing is committed while the cap is P3).

Resource	Deliverable	Effort	Estimated cost (USD)
Boiler inspector or consultant	Review of the design against Texas Health and Safety Code chapter 755 and the ruling	To assess	To assess
Workshop with welder, tube bender and coil former	Prototype built to STR-BLD-001 (P4)	About 2 to 3 weeks of workshop time (estimate, not planned in detail)	2,720 in parts (BOM)
Soil test plot and partner time	P4 soil and steaming tests	To assess with the partners	To assess
Instruments	Flue oxygen analyser, type K probes, scale	Borrowed or bought at P4	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Met (2026-10-25)
P1 Definition	URS, PRS, RSK with hazard analysis, MKT	Met (2026-10-25)
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed, feasibility	Met except the STEM check, which stays open on six plausible rows and two failed rows sent to Amish (S11, B4)
P3 Design outputs	Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability	Met (2026-10-25): D3 model, `cad/fidelity.json`, STR-DWG-002 and 101 to 128, STR-BLD-001, STR-RSK-001
P4 Verification	Test plans and reports, verification report	On hold: TRL cap 3 and the ruling
P5 Validation	Field trials with a partner	On hold
P6 Commercialization	Named deployment partner	On hold

Next steps and decision requested

1. Amish decides #14 (concept: A or B), S2 (hood handling), #12 (restate PR-004 to 4.5 kg/m2), #13 (PR-011 without the hoods), #16 (battery), #17 (PR-016 to PR-018), #18 (bed preparation and hood hold-down), #19 (render site scene), #20 (refeed flash basis) and #21 (firebox skin limit). He may also reverse S1, S3 or S4.
2. The coordinator renders the options picture and the product renders.
3. When Amish chooses, request the written ruling from TDLR (the gate for everything physical).
4. TRL 4 work waits for his go.

References

Every source cited in this repo's documents, Chicago style.

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